Mathematics & Logic Assessment

Question 1: What is the derivative of f(x) = 3x^2 with respect to x?
A. 6x
B. 3x
C. 6x^2
D. 3
Correct Answer: A
Explanation: Applying the power rule: d/dx(3x^2) = 3 * 2x = 6x.
Marks: 2
Topic: Calculus

Question 2: What is the value of 2 raised to the power of 8?
A. 128
B. 256
C. 512
D. 64
Correct Answer: B
Explanation: 2^8 equals 256.
Marks: 2
Topic: Arithmetic

Question 3: In a right-angled triangle, what is the sine of a 90-degree angle?
A. 0
B. 0.5
C. 1
D. Undefined
Correct Answer: C
Explanation: sin(90 degrees) = 1.
Marks: 2
Topic: Trigonometry

Question 4: What is the median of the data set: 3, 7, 9, 12, 15?
A. 7
B. 9
C. 10
D. 12
Correct Answer: B
Explanation: The median is the middle value of an ordered sequence: 9.
Marks: 2
Topic: Statistics